

China Biomedicine Industry

How the industry works, where growth comes from, and what structurally limits it.

Analyst research note · June 2026 · Factual industry overview — not investment advice

Methodology & source discipline

Every factual claim is tied to a named source. The only document held in-project is the 15th Five-Year Plan briefing transcript (State Council Information Office / NDRC), used as the **primary source** for the regulatory section; all other figures are drawn from named external sources (industry analysts, regulators, trade and academic publications) and attributed inline. Inferences are flagged as **(inferred)**. Figures are the most recent available; where a data point predates 2024 it is flagged. Consistent with prior work, this note provides framework and sourced inputs only — it does not give intrinsic-value estimates, entry or stop-loss levels, or any buy/sell recommendation.

#2

world's 2nd-largest
pharma market (2024)

~30%

of the global drug
development pipeline

US\$138bn

record cross-border
out-licensing (2025)

~45%

of new US FDA API
filings (2024)

Sources: IQVIA / Drug Discovery Trends 2025; Jefferies 2025; PharmCube via Pharmsource 2026; USP Quality Matters 2024. API-filing share is new US FDA Drug Master File filings.

1. Industry Purpose & Core Economics

The industry exists to discover, develop, manufacture and sell medicines — increasingly biologic (protein-based) and other complex therapies rather than simple chemical generics. China is now the world's second-largest pharmaceutical market, with sales of about US\$112.8bn in 2024 (IQVIA, via Drug Discovery Trends). On a broader definition, the overall market expanded from RMB1.21tn in 2010 to RMB2.97tn in 2024 (Frontiers in Pharmacology, 2025) [data: 2024].

Value is created at two distinct points. First, in discovery and clinical development, where a molecule that works can be worth billions. Second, in manufacturing and distribution at scale. The economics differ sharply: innovative drugs earn high margins from patents and pricing power, while generics and contract manufacturing are high-volume, thin-margin businesses. The central story of the past five years is China trying to move up this curve — from a generics-and-manufacturing base toward patented innovation.

2. Industry Structure & Competitive Edge

The structure is fragmented and tiered. At the top sit innovators with global ambitions (BeiGene, Hengrui, Akeso, Innovent, 3SBio, CSPC). Below them is a long tail of generic makers, plus a powerful services layer of contract research and manufacturing organisations (CROs / CDMOs) led by WuXi AppTec and WuXi Biologics.

China's competitive edge is speed and cost, not (yet) originality. Lower costs and regulatory flexibility let Chinese start-ups move faster than US counterparts, producing a large inventory of drug prospects (BioPharma Dive, 2025). That edge is now backed by real scientific depth in certain modalities: more than half of the global antibody-drug conjugate (ADC), bispecific-antibody and CAR-T clinical pipeline is China-originated or China-partnered (Evaluate / BioPharma Dive, 2025). (ADCs deliver a toxin directly to tumour cells; bispecifics hit two targets at once; CAR-T re-engineers a patient's own immune cells.)

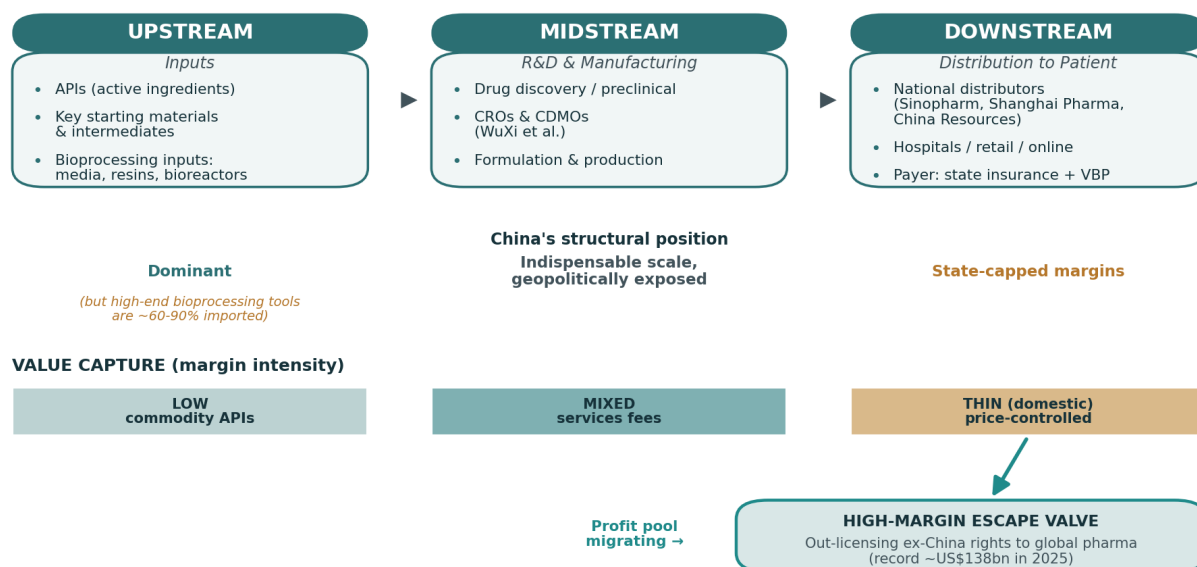
3. Upstream & Downstream (Value Chain)

The value chain runs from raw chemical and biological inputs (upstream), through discovery and manufacturing (midstream), to distribution and the patient (downstream). China's position is strikingly uneven: dominant and cheap at the commoditised bottom, still import-reliant at the high-tech top, and margin-controlled by the state at

the consumer end.

CHINA BIOMEDICINE VALUE CHAIN

Where China dominates, depends, and captures value



Sources: PharmaBoardroom 2024; Blooming 2025-26; USP; IndexBox 2026; BioProcess Online; ResearchAndMarkets; PharmCube via Pharmsource 2026. Stage positions are the author's synthesis.

Figure 1. China biomedicine value chain — position and margin intensity by stage.

Upstream — inputs. China is the world's largest maker of active pharmaceutical ingredients (APIs), the chemical that actually treats disease. Estimates vary by metric: about 20% of global API production by volume (PharmaBoardroom, 2024); chemical-API revenue of RMB578.3bn in 2024, roughly 35.5% of the global API market (Blooming, 2025); and in 2024 China overtook India to hold ~45% of new US API Drug Master File filings (USP, 2024-25) [data: 2024; these measure different things]. The deeper chokehold is one level up: China controls more than 80% of the key starting materials and intermediates market, so 'reshoring' only the final step is ineffective (DrugPatentWatch, 2026). Cost is the moat — Chinese API output runs ~20–30% cheaper than India's (Mantell Associates, 2024).

The exception, and an important constraint, is high-end bioprocessing inputs — cell-culture media, chromatography resins and single-use bioreactor bags used to make biologics. Here China is a net importer: it still imports roughly 60–65% of bioprocessing-grade chromatography media (IndexBox, 2026), and historically about 89% of its cell-culture media was imported [data: 2017; localisation has since risen] (BioProcess Online, 2021). Foreign suppliers (Cytiva, Thermo Fisher, Merck) still lead, as domestic vendors struggle most with hardware and analytics equipment (BioPlan Associates, 2022). **(Inferred:** this is the value-chain equivalent of the chip problem — China owns the cheap bulk inputs but depends on the West for precision tools, a vulnerability if export controls widened.)

Midstream — R&D and manufacturing. The contract-services layer — CROs (research) and CDMOs (development and manufacturing), led by WuXi AppTec and WuXi Biologics, plus innovators' own plants. Its scale is what makes China indispensable to global pharma: a 2024 survey found 79% of biopharma firms relied on at least one Chinese research or manufacturing partner (BIO survey, via van der Kley, 2026). This layer is the most exposed to the US BIOSECURE Act (Section 8).

Downstream — distribution and channels. Finished drugs flow through a wholesale layer dominated by three state-linked national distributors — Sinopharm, Shanghai Pharmaceuticals and China Resources — alongside Jointown and many regional players. The market is still fragmented: there were over 13,000 drug wholesalers as of end-2018 [data: 2018], though policy is consolidating it (ResearchAndMarkets). Drugs reach patients mainly through hospital pharmacies, with retail and a fast-growing online channel taking share (Statista channel data,

2024). The decisive feature is the payer: centralised Volume-Based Procurement (VBP) has manufacturers supply hospitals directly, cutting out the intermediaries that previously marked up prices (Stanford Law, 2025). The state, as dominant buyer, therefore sets the economics of the entire downstream — which is why so much value is now captured by exporting drug rights abroad rather than selling at home.

4. Demand & Growth Drivers

Three structural drivers underpin demand.

- **Demographics and disease burden.** An ageing population is expanding chronic-disease and oncology demand. Healthcare expenditure topped RMB9tn (US\$1.3tn) in 2023, growing ~9.9% a year since 2014 (Swiss Re Institute, via SCMP, 2026) [data: 2023].
- **Global licensing income.** The largest near-term lever is selling Chinese-developed drugs to foreign pharma. Cross-border out-licensing hit a record US\$137.7bn in 2025 — nearly tenfold the 2021 level — and roughly one-third of the industry's total licensing spend involved China-sourced drugs (PharmCube / Jefferies, 2025-26). This is pulled by a Western patent cliff: expirations could strip up to US\$200bn of annual revenue between 2026 and 2030 (Vision Life Sciences, 2026).
- **Domestic volume.** Even with price cuts, reimbursed access to a vast population keeps growing unit volumes.

Selected cross-border out-licensing deals

Chinese licensor	Global partner	Asset / area	Headline value
3SBio	Pfizer	PD-1/VEGF bispecific (SSGJ-707)	up to ~US\$6.0bn*
CSPC	AstraZeneca	Obesity programme	up to US\$18.5bn
GeneQuantum	Biohaven / AimedBio	Antibody-drug conjugates	up to US\$13bn
RemeGen	AbbVie	Oncology	up to US\$5.6bn
Akeso	Summit Therapeutics	PD-1/VEGF bispecific (ivonescimab)	up to US\$5bn†

*Headline values are total potential (upfront plus milestones) as reported by cited press / analyst sources; actual realised value depends on milestones. *3SBio–Pfizer included a ~US\$1.25bn upfront. †Akeso–Summit signed in Dec 2022; included here as the deal that opened the licensing wave. Sources: SCMP 2026; Pharmasource (PharmCube) 2026; BioPharmaAPAC 2026 [data: 2022–2026].*

5. Supply Side, Cost Structure & Constraints

China's supply advantage is a deep, low-cost base of scientists, clinical-trial patients and manufacturing; a 2024 survey found 79% of biopharma firms relied on at least one Chinese partner (BIO survey, 2026). The binding domestic constraint is price. Through Volume-Based Procurement — centralised bulk buying that trades guaranteed volume for steep discounts — the first nine rounds delivered average price cuts of about 50%, with some products down more than 90% (NHSA, via Stanford Law, 2025). The effect is a margin squeeze: price negotiation and procurement have severely compressed margins, especially for small and medium generic makers (Frontiers in Pharmacology, 2025). VBP is also a consolidation engine, rewarding low-cost players with scale and forcing weaker ones out (inferred from evidence that procurement raised market concentration).

6. Technology Changes

The centre of gravity has shifted from generics to advanced biologics. Oncology is roughly 62% of the R&D pipeline (DrugPatentWatch, 2025), concentrated in ADCs, bispecifics and cell therapies. A newer strength is targeted protein degradation (a way to 'delete' disease-causing proteins once thought undruggable): China now accounts for ~38% of global TPD publications and ~37% of patents (Clarivate, 2025).

AI-assisted discovery is emerging but unproven at scale — China's 'AI + Pharma' ecosystem had grown to 98 companies by 2024 [data: 2024] (Frontiers in Pharmacology, 2025). A persistent technical limit is data

acceptance abroad: drugs tested only in China can be rejected by Western regulators, as when the global trial of the bispecific ivonescimab hit progression-free survival but missed on overall survival (BioPharmaAPAC, 2026) — exposing the risk that China-only data does not always travel.

7. Regulatory Changes (15th Five-Year Plan focus)

The 15th Five-Year Plan transcript treats biomedicine as a designated growth priority rather than a mature sector. Three primary-source references stand out:

[15th Five-Year Plan transcript, p.13]

The plan names fostering “emerging pillar industries, including integrated circuits, biomedicine, and aerospace” — placing biomedicine alongside chips as a strategic pillar.

[15th Five-Year Plan transcript, p.15]

It targets “decisive breakthroughs in key and core technologies” in fields including “bio-manufacturing” — signalling continued state push into biologic production capability.

[15th Five-Year Plan transcript, p.20]

In the marine chapter it cites fostering emerging industries “such as marine biomedicine” — a narrower, specific sub-sector mention.

Beyond the Plan, the operative shift is reimbursement reform that rewards innovation. On 7 December 2025 China released its updated National Reimbursement Drug List (114 new drugs, 50 of them Class 1 innovative) and its first-ever Commercial Health Insurance Innovative Drug List of 19 high-value drugs, both effective 1 January 2026 (CMS Law / Pharmaceutical Technology, 2025-26). Notably, the NHSA did not disclose an average price-reduction rate for newly listed drugs, possibly a turning point where cost-cutting is weighted less heavily (Pharmaceutical Technology, 2026). Upstream, a March 2025 State Council directive set a goal of turning China from a major pharmaceutical manufacturer into a pharmaceutical powerhouse by 2027 (NMPA, 2025).

8. Structural Changes: Export Markets & the Supply Glut

Export markets. The key new risk is the US BIOSECURE Act, signed into law on 18 December 2025 as part of the FY26 defence bill. It restricts US federal procurement and grants involving ‘biotechnology companies of concern’, defined via the Department of Defense’s 1260H list (Latham & Watkins; Arnold & Porter, 2025). Prominent CRO/CDMO firms WuXi AppTec and WuXi Biologics are not currently subject to the restrictions, but a five-year safe harbour applies to existing contracts if they are later designated. The drug-licensing channel itself appears, for now, politically tolerated.

The supply glut. China’s ‘involution’ problem — too many firms chasing the same target — is acute in oncology. As of mid-2025, 23 PD-1/PD-L1 products had been approved in China, more than double the 11 in the EU, and procurement and reimbursement have forced price cuts averaging 58–80% (over 90% in some rounds), pushing firms overseas (CIRSD, 2026). The structural release valves are therefore: (a) out-licensing to global partners; (b) policy-driven consolidation via VBP; and (c) the new commercial-insurance lane that lets high-value drugs avoid the deepest cuts. None has yet fully cleared the glut (inferred from the continued pace of new domestic entrants; CKGSB, 2025).

9. Medium-Term Outlook (5–10 years)

Forecasts bracket a range: Statista projects the pharmaceuticals market growing ~5.5% a year to about US\$164.6bn by 2030, with biologics growing faster. Continued upgrading toward innovation is the consensus direction, but margins, geopolitics and the glut temper it. Independent analysts frame this as a maturing from rapid expansion to ‘Innovation 2.0’ — sustainable, quality-focused development (Clarivate, 2025). The realistic base case (inferred synthesis): strong global licensing income and rising domestic volume, offset by thin domestic pricing and recurring geopolitical friction.

INVESTOR SYNTHESIS

Core economic engine

Two engines, not one: a low-margin, high-volume domestic manufacturing/generics base, and a high-margin innovation layer whose cash increasingly comes from licensing IP to global pharma rather than selling drugs at home.

Primary growth lever

Out-licensing. Patent-cliff-driven demand from multinationals turned Chinese R&D into a ~US\$138bn export of drug rights in 2025 — the single biggest swing factor in the bull case.

Structural constraints investors underestimate

Domestic pricing. VBP and reimbursement negotiation cap value capture inside China — the country is over 30% of the global pipeline yet only ~3–5% of global drug sales; the new commercial-insurance channel is real but small (~3.3% of health spend). There is also an upstream tool dependency in high-end bioprocessing equipment.

Key risk that could change trajectory

Geopolitics reaching the licensing channel. BIOSECURE so far targets services (CDMOs), not licensing — any extension to in-licensing, or pressure on China-only clinical-data acceptance, would hit the primary growth lever directly.

What kind of companies tend to win

(Inferred from the structure above.) Innovators with differentiated, globally credible assets in hot modalities (ADCs, bispecifics, cell therapy) that command real licensing economics; low-cost manufacturers and CDMOs with the scale to survive VBP consolidation; and firms generating data to multi-region (not China-only) standards. Crowded 'me-too' players in saturated targets like PD-1 are structurally disadvantaged.

Selected sources

1. 15th Five-Year Plan briefing transcript, State Council Information Office / NDRC (primary source, in-project).
2. Frontiers in Pharmacology / PMC, "Technological and industrial trends in China's pharmaceutical sector," Sept 2025.
3. Drug Discovery Trends, "China's biopharma sector enters 'Innovation 2.0' era" (IQVIA / Clarivate data), Dec 2025.
4. Statista, "Pharmaceuticals – China" market forecast, 2025.
5. Pharmasource (PharmCube data), "China biopharma out-licensing surges to record \$137.7bn in 2025," Feb 2026.
6. South China Morning Post, "Chinese drug makers strike record US\$136bn out-licensing deals in 2025," Jan 2026.
7. Fierce Biotech / BioPharma Dive, Jefferies equity research on China out-licensing, 2025.
8. Vision Life Sciences, "China Biotech Outbound Licensing Tracker," Feb 2026.
9. BioPharmaAPAC, "Innovation Powerhouse or Inflated Bubble? Reading China's First-in-Class Boom," 2026.
10. ScienceDirect, "Impacts of China's National Volume-Based Procurement policy," 2025.
11. Stanford Law School, Law & Biosciences Blog, "Drug Centralized Procurement in China," Feb 2025.
12. Simon-Kucher, "What the latest volume-based procurement changes in China mean for pharma," Jul 2025.
13. CIRSD, "A Biopharmaceutical Superpower: China's Rise, Its Limits, and What Comes Next," May 2026.
14. CKGSB, "China's Biotech Rise and Global Innovation Challenges," Oct 2025.
15. NMPA / State Council, "Opinions on Comprehensively Deepening the Reform of Drug and Medical Device Regulation," Mar 2025.
16. CMS Law & Pharmaceutical Technology, 2025 NRDL and first Commercial Insurance Innovative Drug List, Dec 2025–Jan 2026.
17. South China Morning Post / Swiss Re Institute, China health-insurance and expenditure data, Mar 2026.
18. Latham & Watkins; Arnold & Porter; Foley Hoag, BIOSECURE Act becomes law analyses, Dec 2025.
19. USP Quality Matters, "Global manufacturing capacity for APIs remains concentrated," 2024–25.
20. Blooming, "China's API Industry Enters a New Era" & "China API Exports Enter 3.0 Era," 2025–26.
21. PharmaBoardroom, "China Pharmaceutical Market Report 2024."
22. IndexBox, "Asia Cellulose-Based Chromatography Media Market," 2026; BioProcess Online / BioPlan Associates, 2021–22.
23. ResearchAndMarkets / PRNewswire, "China Pharmaceutical Distribution Industry Report."
24. DrugPatentWatch, "The role of China in the global generic drug API market," 2026.